

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 3 Winter 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: 25AF1245PC303 - DISCRETE MATHEMATICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Power set of empty set has exactlySubsets.
A.One B.Two C.Zero D.Three
- 2 A binary operation Δ on a set of integers is defined as $x\Delta y = x + y$ which one of the following statement is true about?
A. commutative but not associative B. Both commutative and associative
C. associative but not commutative D. neither commutative nor associative
- 3 Let A be finite set having x elements and B be a finite set having y elements. What is the number of distinct functions mapping B into A.
A.x B.2 C.y D. $y!/(y-x)!$
- 4 Which of the following is NOT a proposition?
A. 7 is a prime number B. $x + 3 = 10$
C. Mumbai is the capital of Maharashtra D. $2 + 2 = 4$
- 5 Which of the following is logically equivalent to $\neg(P \wedge Q)$?
A. $\neg P \wedge \neg Q$ B. $\neg P \vee \neg Q$ C. $P \vee Q$ D. $\neg P \rightarrow \neg Q$
- 6 The negation of the statement $\forall x(P(x))$ is:
A. $\forall x(\neg P(x))$ B. $\exists x(\neg P(x))$ C. $\neg \exists x(P(x))$ D. $\exists x(P(x))$
- 7 Which of the following statements is NOT equivalent to $\forall x \exists y P(x, y)$?
A. For every x, there exists a y such that $P(x, y)$
B. $\exists y \forall x P(x, y)$
C. For each x, some y satisfies $P(x, y)$
D. Every x has at least one y with property P
- 8 Which of the following structures must have an identity and inverses?
A. Semigroup B. Monoid C. Group D. Ring
- 9 Which algorithm is used to find a minimum spanning tree?
A. Breadth First Search B. Depth First Search
C. Dijkstra's Algorithm D. Kruskal's Algorithm

- 10 A connected graph has an Euler circuit if:
- All vertices have odd degree
 - Exactly two vertices have odd degree
 - All vertices have even degree
 - The graph is complete
- 11 A graph that has no cycles is called:
- Complete graph
 - Bipartite graph
 - Tree
 - Planar graph
- 12 Which rule of inference is used in the argument: If $P \rightarrow Q$ and P is true, then Q is true.
- Modus Tollens
 - Hypothetical Syllogism
 - Modus Ponens
 - Resolution

Q2. Solve the following.

- A) Translate each of these statements into logical expressions using predicates, 6 quantifiers, and logical connectives. Let $T(x)$ = x is a tool, $C(x)$ = x is in the correct place, and $E(x)$ = x is in excellent condition.
- All tools are in the correct place and are in excellent condition.
 - Nothing is in the correct place and is in excellent condition.
 - One of your tools is not in correct place, but it is in excellent condition.
- B) Show that the argument form with premises $(p \wedge t) \rightarrow (r \vee s)$, $q \rightarrow (u \wedge t)$, $u \rightarrow 6$ p , and $\neg s$ and conclusion $q \rightarrow r$ is valid by using rules of inference.

Q3. Solve the following.

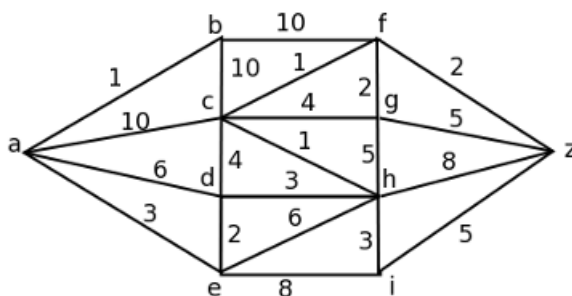
- A) Show that $A + B = (A \cup B) \cap (\sim A \cup \sim B)$ 6
- B) Given $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. Let xRy whenever y is divisible by x . Find and 6 Write R as set of ordered pairs. Is R so found reflexive, symmetric and transitive? Is R Equivalence? Justify with a valid argument.

Q4. Solve Any Two of the following.

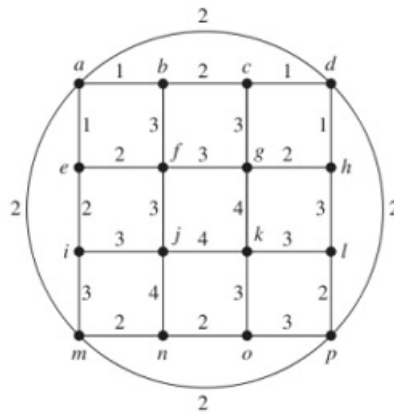
- A) It is given that white tiger population of Orissa (INDIA) is 30 at time $n = 0$ and 326 at time $n = 1$. Also the increase from time $n - 1$ to time n is twice the increase from time $n - 2$ to $n - 1$. Write and solve recurrence relation for growth rate of tiger.
- B) For the difference equation given below find particular solution. 6
- $$a_r - 4a_{r-1} + 4a_{r-2} = (r + 1)2^r$$
- C) Solve the following non-homogeneous recurrence relation $a_n = 2a_{n-1} + 3(2^n)$ 6 where, $a_0 = 3$.

Q5. Solve Any Two of the following.

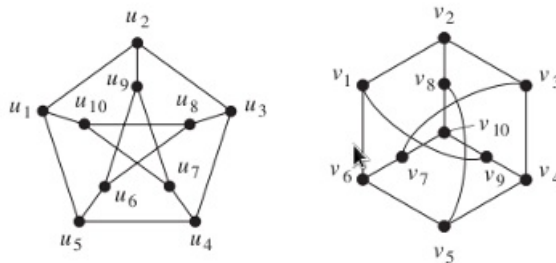
- A) Apply the Dijkstra's Algorithm to find the length of a shortest path between **a** and 6 **z** in the given weighted graph.



- B) Use Kruskal's algorithm to find a minimum spanning tree for the weighted graph in 6 figure given below.



- C) With justification determine whether the given pair of graphs is isomorphic or not. 6



Q6. Solve Any Two of the following.

- A) Let G be a reduced residue system modulo 15, say, $G = \{1, 2, 4, 7, 8, 11, 13, 14\}$. 6
- Find Multiplication table of G .
 - Find $2^{-1}, 7^{-1}$.
 - Find the orders and subgroups generated by 2, 7, and 11.
- B) Define the following Terms: 6
- Abelian Group
 - Identity Element
 - Inverse Element
 - Group
 - Monoid
 - Rings
- C) Consider the set N of positive integers, and let $*$ denote the operation on least 6 common multiple (lcm) on N .
- Find $4 * 6$ and $3 * 5$.
 - Is $(N, *)$ is a semigroup? Is it commutative?
 - Which element in N , if any, have inverse and what are they?

*** End ***